Complex Numbers: Complete Solution Sets
Answer Key

Algebra 2 Complex Numbers

Quick Answers

1. $11 + 10i$
2. $-i$
3. $7i$, $-7i$
4. $3 + 4i$, $3 - 4i$
5. 0
6. 2, -2, $2i$, $-2i$
7. $-2 + 3i$, $-2 - 3i$
8. 0, $2i$, $-2i$
9. 0
10. 1, -1, $2i$, $-2i$
11. $\frac{1}{2} + i$, $\frac{1}{2} - i$
12. 1, $3i$, $-3i$

Curriculum

Level: Algebra 2 Complex Numbers
HSN-CN.A.1–A.3, HSN-CN.B.4–B.6 – *problems 1, 2*
HSN-CN.C.7–C.9 – *problems 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

9. *If they got 32: squared the binomial with i squared taken as plus 1, which turns the substitution into a nonzero real number*

1. Multiply $(2 + 3i)(4 - i)$ and write the result as $a + bi$.

Step 1. Distribute exactly as with binomials — four products: $2 \cdot 4 = 8$, $2 \cdot (-i) = -2i$, $3i \cdot 4 = 12i$, $3i \cdot (-i) = -3i^2$.
Step 2. Replace i^2 by -1: $-3i^2 = -3(-1) = 3$. This is the only step where an imaginary product becomes real.
Step 3. Collect real parts $8 + 3 = 11$ and imaginary parts $-2i + 12i = 10i$. $\boxed{11 + 10i}$
Common wrong answer: $5 + 10i$ — from treating i^2 as $+1$, which subtracts the 3 instead of adding it.

2. Simplify i^{23}.

Step 1. Powers of i repeat every four steps: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, and then the cycle starts over. So only the remainder of the exponent on division by four matters.
Step 2. $23 = 4 \cdot 5 + 3$, so $i^{23} = (i^4)^5 \cdot i^3 = 1 \cdot i^3$.
Step 3. $i^3 = i^2 \cdot i = -i$. $\boxed{-i}$
Watch the period: it is four, not two — “the exponent is odd, so the answer is i” gets this one backwards.

3. Solve $x^2 + 49 = 0$; report the complete solution set.

Step 1. Nothing to factor over the reals, and the equation is already a square set equal to a number: isolate the square. $x^2 = -49$.
Step 2. Take square roots of both sides, keeping BOTH signs: $x = \pm\sqrt{-49} = \pm\sqrt{49}\sqrt{-1} = \pm 7i$.
Step 3. A degree-2 equation has two solutions, and here they are a conjugate pair.

$\boxed{x = 7i \text{ or } x = -7i}$

Check: $(7i)^2 + 49 = 49i^2 + 49 = -49 + 49 = 0$. The same arithmetic works for $-7i$, because squaring kills the sign.

4. Solve $(x-3)^2 + 16 = 0$; report the complete solution set.

Step 1. The variable appears only inside one square, so square roots beat the quadratic formula here. Isolate: $(x-3)^2 = -16$.
Step 2. $x - 3 = \pm\sqrt{-16} = \pm 4i$.
Step 3. Add 3 to both sides. The real part comes from the shift, the imaginary part from the square root. $\boxed{x = 3 + 4i \text{ or } x = 3 - 4i}$
Check: $(3 + 4i - 3)^2 + 16 = (4i)^2 + 16 = -16 + 16 = 0$.

5. Show $x = 1 + 4i$ solves $x^2 - 2x + 17 = 0$.

Step 1. "Is it a solution?" is answered by substitution: replace every x and simplify to one number. A solution makes the left side 0.
Step 2. Square first: $(1+4i)^2 = 1 + 8i + 16i^2 = 1 + 8i - 16 = -15 + 8i$.
Step 3. Now the linear term: $-2(1+4i) = -2 - 8i$.
Step 4. Add the three pieces: $(-15+8i) + (-2-8i) + 17 = (-15-2+17) + (8-8)i$. $\boxed{0}$
Because the substitution gives exactly 0, $x = 1 + 4i$ is a solution.
Common wrong answer: 32 — from squaring with i^2 taken as $+1$, which makes the imaginary parts cancel into a nonzero real number.

6. Every zero of $p(x) = x^4 - 16$.

Step 1. Degree 4, so expect four zeros. The expression is a difference of squares, so factor rather than expand anything.
Step 2. $x^4 - 16 = (x^2 - 4)(x^2 + 4)$, and the first factor is a difference of squares again: $(x-2)(x+2)(x^2+4)$.
Step 3. Set each factor to zero. $x - 2 = 0$ and $x + 2 = 0$ give the real zeros; $x^2 + 4 = 0$ gives $x^2 = -4$, so $x = \pm 2i$.
Step 4. Four zeros, as the degree promised. $\boxed{x = 2,\ -2,\ 2i,\ -2i}$
Note: $x^2 + 4$ does not factor over the reals, which is exactly why two of the zeros are non-real.

7. Solve $x^2 + 4x + 13 = 0$; report the complete solution set.

Step 1. No integer pair multiplies to 13 and adds to 4, so the quadratic formula is the tool: $a = 1$, $b = 4$, $c = 13$.
Step 2. Discriminant: $b^2 - 4ac = 16 - 52 = -36$. Negative, so both solutions are non-real.
Step 3. $\sqrt{-36} = 6i$, so $x = \dfrac{-4 \pm 6i}{2} = -2 \pm 3i$.
Step 4. The pair is a conjugate pair, as it must be for a quadratic with real coefficients.

$\boxed{x = -2 + 3i \text{ or } x = -2 - 3i}$

Check the real part: the two solutions must average to $-b/(2a) = -2$, and they do.

8. Complete zero set of $p(x) = x^3 + 4x$.

Step 1. Degree 3 promises three zeros. Every term has a factor of x, so pull it out before anything else — that is what keeps $x = 0$ from being lost.
Step 2. $x^3 + 4x = x(x^2 + 4)$.
Step 3. $x = 0$ from the first factor. From the second, $x^2 = -4$, so $x = \pm 2i$.
Step 4. Three zeros for a cubic. $\boxed{x = 0,\ 2i,\ -2i}$
Common wrong answer: dividing both sides by x at Step 2 destroys the solution $x = 0$ and leaves only two zeros for a cubic.

9. Ben substituted $x = -3 - 4i$ into $x^2 + 6x + 25$ and got 32. Redo it.

Step 1. Substitution is only as good as the squaring step, so do $(-3 - 4i)^2$ carefully and keep $i^2 = -1$.
Step 2. $(-3 - 4i)^2 = 9 + 24i + 16i^2 = 9 + 24i - 16 = -7 + 24i$.
Step 3. Linear term: $6(-3 - 4i) = -18 - 24i$.
Step 4. Add: $(-7 + 24i) + (-18 - 24i) + 25 = (-7 - 18 + 25) + (24 - 24)i$. $\boxed{0}$
Ben's mistake: he treated i^2 as $+1$, so his squaring gave $9 + 24i + 16 = 25 + 24i$; that turns the whole substitution into 32 instead of 0. Since the correct value is 0, $-3 - 4i$ *is* a solution — the conjugate partner of $-3 + 4i$.

10. Complete solution set of $x^4 + 3x^2 - 4 = 0$.

Step 1. Only even powers of x appear, so treat it as a quadratic in x^2 and factor in that variable first.
Step 2. $x^4 + 3x^2 - 4 = (x^2 + 4)(x^2 - 1)$, because $4 \cdot (-1) = -4$ and $4 + (-1) = 3$.
Step 3. $x^2 - 1 = 0$ gives the real solutions $x = 1$ and $x = -1$. $x^2 + 4 = 0$ gives $x^2 = -4$, so $x = \pm 2i$.
Step 4. Four solutions for a degree-4 equation. $\boxed{x = 1,\ -1,\ 2i,\ -2i}$
Watch out: stopping at $x^2 = 1$ and $x^2 = -4$ is only half the work — each of those is still a quadratic and contributes two solutions.

11. Solve $4x^2 - 4x + 5 = 0$; report the complete solution set.

Step 1. The leading coefficient is not 1 and nothing factors, so use the quadratic formula with $a = 4$, $b = -4$, $c = 5$.
Step 2. Discriminant: $16 - 80 = -64$, and $\sqrt{-64} = 8i$.
Step 3. $x = \dfrac{4 \pm 8i}{8}$. Divide each part of the numerator by 8: the real part is $4/8$ and the imaginary part is $8i/8$.
Step 4. In standard form the real part is a fraction and the imaginary part is 1.

$$\boxed{x = \tfrac{1}{2} + i \text{ or } x = \tfrac{1}{2} - i}$$

Common wrong answer: $\frac{1}{2} \pm 8i$ — the 8 in the numerator was divided but the $8i$ was not.

12. Synthesis: $x = 3i$ **is a zero of** $p(x) = x^4 - 2x^3 + 10x^2 - 18x + 9$. **Find the complete solution set.**

Step 1. p has real coefficients, so non-real zeros come in conjugate pairs: $x = -3i$ is a zero too. Two zeros in hand means a known quadratic factor.
Step 2. That factor is $(x - 3i)(x + 3i) = x^2 - 9i^2 = x^2 + 9$.
Step 3. Divide: $x^4 - 2x^3 + 10x^2 - 18x + 9 = (x^2 + 9)(x^2 - 2x + 1)$. (Check the ends: $x^2 \cdot x^2 = x^4$ and $9 \cdot 1 = 9$.)
Step 4. $x^2 - 2x + 1 = (x - 1)^2$, so $x = 1$ is a zero of multiplicity two.
Step 5. Counted with multiplicity the degree-4 polynomial has its four zeros: 1 twice, and the conjugate pair. $\boxed{x = 1 \text{ (double)},\ 3i,\ -3i}$

Why the conjugate is forced: if only $3i$ were a zero, the factor $x - 3i$ would leave a factor with non-real coefficients, which a polynomial with integer coefficients cannot have.